

Midterm 1 will cover topics from 4.10 (in Vol. 1), 1.1, 1.2, 1.3, 1.5, and 3.1

- Problem 1 and Problem 2 - Riemann sum computation with specified n .
- Problem 3 and 4 - limits of Riemann sum set-up without evaluating.
- Problem 5 and Problem 6 - evaluate limit of Riemann sums and recognize them as definite integrals.
- Problem 7 - evaluate definite integral using the geometric meaning (i.e. net area) of definite integrals.
- Problem 8 - find derivatives by applying fundamental theorem of calculus Part 1.
- Problem 9 - indefinite and definite integrals.

1. Consider the function $f(x) = x^2 - 2$ on the interval $[2, 8]$.

- 1). Sketch the graph of the function on the given interval.
- 2). Illustrate the left Riemann sum on the given interval with $n = 3$ by sketching the appropriate rectangles on the graph of the function.
- 3). Calculate the left Riemann sum on the given interval with $n = 3$.

2. Consider the function $f(x) = -x^2 + 3$ on the interval on $[-1, 3]$.

- 1). Sketch the graph of the function on the given interval.
- 2). Illustrate the right Riemann sum on the given interval with $n = 4$ by sketching the appropriate rectangles on the graph of the function.
- 3). Calculate the right Riemann sum on the given interval with $n = 4$.

3. Consider the function $f(x) = 3 + \ln x$ on $[0, 4]$.

- 1). If the interval $[0, 4]$ is divided into n subintervals of equal length, what is the width of the each subinterval?
- 2). Find the right endpoint x_i of the i th subinterval.
- 3). Express $\int_0^4 (3 + \ln x) dx$ as a limit of Riemann sums with right endpoints. *Note: You do not need to simplify and evaluate the limit.*

4. Consider the function $f(x) = x\sqrt{x+2}$ on $[1, 3]$.

- 1). If the interval $[1, 3]$ is divided into n subintervals of equal length, what is the width of the each subinterval?
- 2). Find the right endpoint x_i of the i th subinterval.
- 3). Express $\int_1^3 x\sqrt{x+2} dx$ as a limit of Riemann sums with right endpoints. *Note: You do not need to simplify and evaluate the limit.*

5. 1). Simplify the sum $R_n = \sum_{i=1}^n \frac{2}{n} \left(2 - \frac{2i}{n} + \frac{8i^3}{n^3} \right)$. Write your answer as a function of n without any summation signs.

2). Use your result from Part 1), evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left(2 - \frac{2i}{n} + \frac{8i^3}{n^3} \right)$.

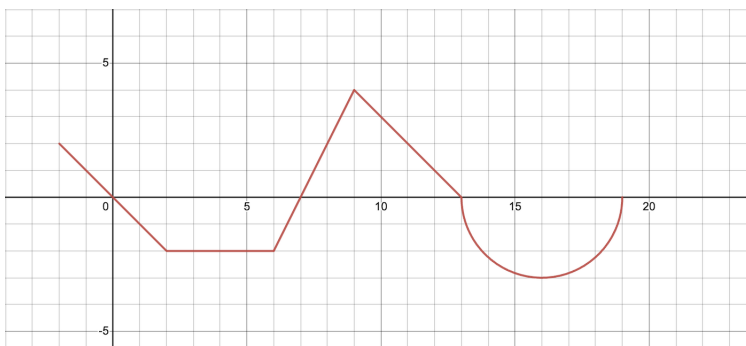
3). Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left(2 - \frac{2i}{n} + \frac{8i^3}{n^3} \right)$ as a definite integral: $\int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{2cm}} dx$.

6. 1). Simplify the sum $R_n = \sum_{i=1}^n \frac{1}{n} \left(\frac{2i}{n} + \frac{i^2}{n^2} \right)$. Write your answer as a function of n without any summation signs.

2). Use your result from Part 1), evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \left(\frac{2i}{n} + \frac{i^2}{n^2} \right)$.

3). Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \left(\frac{2i}{n} + \frac{i^2}{n^2} \right)$ as a definite integral: $\int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{2cm}} dx$.

7. Consider the $f(x)$, whose graph is given as follows. Evaluate the following integrals.



1) $\int_{-2}^2 f(x) dx$

2) $\int_0^7 f(x) dx$

3) $\int_6^9 f(x) dx$

4). $\int_7^{16} f(x) dx$

5). $\int_9^{19} f(x) dx$

6). $\int_{-2}^{19} f(x) dx$

8. Find the derivative $F'(x)$ of each of the following functions.

1). $F(x) = \int_1^x t^3 dt$

2). $F(x) = \int_5^{\sqrt{x}} \frac{\cos t^2}{t^4} dt$

3). $F(x) = \int_{3x}^{-5} \sin(t^2) dt$

4). $F(x) = \int_{\ln x}^{e^x} 7 \sin t dt$

9. Find the following integrals.

1). $\int (-4 \sin x + 6 \cos x) dx$

3). $\int_{-e}^{-1} \frac{10}{x} dx$

5). $\int \frac{x^5 \sqrt{x} + 1}{x^4} dx$

7). $\int_1^6 \frac{3}{2x-1} dx$

9). $\int_0^{\pi/6} \sec^2(2x) dx$

11). $\int \frac{\cos x}{1 + \sin^2 x} dx$

13). $\int \frac{x}{(1 - 2x^2)^5} dx$

15). $\int_1^e \frac{(\ln x)^2}{x} dx$

17). $\int_0^{\pi/2} \frac{\sin x}{1 + \cos x} dx$

19). $\int_0^1 \frac{(\tan^{-1} x)^3}{1 + x^2} dx$

21). $\int x e^{-2x} dx$

23). $\int x \cos(4x) dx$

2). $\int \frac{5 - 2xe^x}{x} dx$

4). $\int e^{-3x+2} dx$

6). $\int_1^4 \frac{2x^2 + 2x - \sqrt{x}}{\sqrt{x}} dx$

8). $\int_1^{\sqrt{3}} \frac{3}{1 + x^2} dx$

10). $\int_0^{\pi/3} \cos^4 \theta \sin \theta d\theta$

12). $\int x^2 e^{-4x^3} dx$

14). $\int \frac{(\sin^{-1} x)^8}{\sqrt{1 - x^2}} dx$

16). $\int_1^2 \frac{e^{1/x}}{x^2} dx$

18). $\int \frac{(\sqrt{x} + 3)^3}{\sqrt{x}} dx$

20). $\int x^2 \ln x dx$

22). $\int x \sin(2x) dx$

24). $\int \frac{\ln x}{\sqrt{x}} dx$

Formula Sheet

$$\sum_{i=1}^n c = cn$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6}$$

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^4 + 2n^3 + n^2}{4}$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$